

CARTLY

Sprint 1 & 2 Update and Team Contributions

Autonomous Customer-Support Resolution Platform · Capstone Project 01
July 2026 · Avishka Jindal, Hiten Mistry, Yash Parmar

1. Sprint Status Overview

Sprint	Focus	Status	Notes
Sprint 0	Discover and Define	DONE	All five exit criteria met.
Sprint 1	Design and Derisk	DONE*	*Golden set locked at 10 placeholder examples, not the full set — carried into Sprint 2.
Sprint 2	Build the Core	IN PROGRESS	Substantial extra design/doc work done ahead of the core build; core build itself not yet started.
Sprint 3	Harden, Scale and Optimise	NOT STARTED	—
Sprint 4	Verify, Operate and Present	NOT STARTED	—

2. Sprint 1 Updates — Design and Derisk

STATUS: DONE

Exit Criterion	Status	Evidence
Architecture diagram and topology spec	MET	C4 L1/L2, HLD
Orchestration decision record	MET	PDLC v1.1 Addendum (Stage 3)
LLM gateway and model routing policy design	MET	Build Tech Stack v1.0, §LiteLLM
Agent registry design	MET	LLD, agent topology
Synthetic pipeline v1	MET	6-category pipeline design
Evaluation harness and golden set locked	PARTIAL	Harness locked; golden set is a 10-example placeholder, not the 500+200 set
Working spike on INR 500 threshold assumption	MET	12-scenario spike, 12/12 pass

One item carried forward rather than silently closed: the golden set locked in Sprint 1 is a 10-example placeholder set (SHA256-hashed manifest), not the full 500 Bitext + 200 synthetic target. Tracked as open work in Sprint 2.

3. Sprint 2 Updates — Build the Core

STATUS: IN PROGRESS

Official Sprint 2 exit criteria (per PDLC v1.0)

Exit Criterion	Status	Evidence
Core agent path working end to end (Triage, Order, Refund)	NOT STARTED	—
LLM gateway live with LiteLLM	NOT STARTED	Config designed, not deployed
Vector store loaded with policy documents	NOT STARTED	Qdrant Cloud (supersedes v1.0's ChromaDB reference)
First RAGAS evaluation and cost baseline measured	NOT STARTED	—

Extra work already completed (beyond Sprint 0 / Sprint 1 baseline scope)

Not required to close Sprint 0 or Sprint 1, but done and sitting in the project repository — de-risks the Sprint 2 core build and pulls forward parts of Sprint 3's design surface.

- PDLC v1.1 Addendum — ingestion/worker split, MCP tool layer, revised data stores, Stage 3.9–3.11 agent behaviours.
- PDLC v1.2 Addendum — Damage Assessment Agent, vision pipeline, three-way branch.
- Build Tech Stack v1.0 — full 16-category stack locked with rationale.
- Cartly_Policy_KB_v1.1.md and four customer-facing policy documents.
- POC codebase (cartly_poc/) — runnable refund-eligibility slice.
- C4 diagrams (L1/L2), HLD, LLD, and full sequence diagram set (happy + unhappy paths).
- R/M/H edge-walk and the A1–A14 assumptions register.
- PRD + risk register reconciliation v1.1.

Still open to close Sprint 2

- model_routing.yaml — not yet written.
- Full project scaffold — not yet built.
- Updated C4 diagram incorporating the Damage Assessment Agent.
- Golden Set — 500 Bitext + 200 synthetic tickets (carried from Sprint 1).
- Synthetic data pipeline update — 7th category, Emotional Customer Variants (800 examples).
- README, runbook, evaluation report.

4. Latest Project Progress

Sprint 0 and Sprint 1 are closed. The team is mid-Sprint-2: the official Sprint 2 build (live agent path, LiteLLM gateway, Qdrant-backed retrieval, first RAGAS baseline) has not started yet, but the design and documentation surface it depends on is substantially ahead of schedule — architecture, policy content, POC code, diagrams, and analytical frameworks are all in place.

The locked stack (source of truth: PDLC v1.2 + Build Tech Stack v1.0) is: LangGraph orchestration, Qdrant Cloud vector store, bge-small-en-v1.5 embeddings, Supabase (Postgres + Auth + Storage), Upstash Redis, React + Vite frontend, LiteLLM virtual model tiers, FastMCP tool server, Langfuse observability, Llama Guard safety

classification, and RAGAS/DSPy/promptfoo for evaluation. An earlier exploratory branch (ChromaDB / bge-large / SQLite / NVIDIA NIM / Next.js) was evaluated and explicitly rejected.

Immediate next step: start the real build (not the POC scaffold) using Claude Code, gated on writing `model_routing.yaml` and standing up the project scaffold.

5. Collective Team Contributions

Work completed together across Sprints 0–1 and the extra Sprint 2 groundwork:

- Product Development Lifecycle taken from v1.0 through v1.1 and v1.2 — discovery, design, and the Damage Assessment Agent addendum.
- Full technology stack evaluated and locked across 16 categories, with alternatives considered and rejected on the record.
- Complete architecture documentation: C4 (L1/L2), HLD, LLD, sequence diagrams for both happy and unhappy paths.
- Policy content: an internal, audited Policy KB plus four customer-facing policy documents.
- A runnable proof-of-concept for the refund-eligibility slice, with INR-denominated cost tracking.
- Formal analytical frameworks: the R/M/H edge-walk and the A1–A14 assumptions register.
- PRD and risk register reconciled and brought into alignment (v1.1).
- A working spike validating the INR 500 autonomous refund threshold (Assumption A2), and a first (placeholder) golden set with a locked, hashed manifest.

6. Individual Contributions

Attribution follows the three-track split proposed at Sprint 1 planning — Avishka: architecture and diagrams; Hiten: orchestrator and refund/agent implementation; Yash: safety critic, evaluation, and demo. All work completed since Sprint 1 (the “extras” in Section 3) has been divided evenly across the three tracks, four artifacts each, following that same split.

Avishka Jindal — Architecture and Diagrams

Artifact	Contribution
PDLC v1.1 Addendum	Stage 3 redesign: ingestion/worker split, MCP tool-access layer, revised data-store choices, formalized Stage 3.9–3.11 agent behaviours.
Build Tech Stack v1.0	Full 16-category stack lock with rationale and rejected alternatives per category.
C4 diagrams (L1/L2), HLD, LLD	System context and container diagrams, high-level design, low-level design for core agents.
Sequence diagrams — happy path	Tier 1, Tier 2, and Tier 3–4 resolution flows.

Hiten Mistry — Orchestrator and Refund/Agent Implementation

Artifact	Contribution
PDLC v1.2 Addendum	Damage Assessment Agent merged into the locked stack; vision pipeline three-way branch (auto-approve / human-review / fraud-flag); fraud queue as a <code>queue_type</code> tag on the handoff table.
POC codebase (cartly_poc/)	8-file, runnable refund-eligibility slice; INR-denominated cost output.

Artifact	Contribution
Cartly_Policy_KB_v1.1.md	Internal, engineering-facing policy knowledge base; audited, two PDLC §4.2 drifts corrected.
Four customer-facing policy docs	Return & Refund, Damage & Defect, Cancellation, Purchase Protection.

Yash Parmar — Safety Critic, Evaluation, and Demo

Artifact	Contribution
Sequence diagrams — unhappy paths	Safety Critic retry exhaustion → escalation (bounded Reflexion); kill-switch bypass → human queue; Tier 2 human reviewer rejection.
R/M/H edge-walk	Every decision point in the system classified as Rule-based, Model-based, or Human-decided.
Assumptions register, A1–A14	Formally numbered assumptions with rationale and confirmation path.
PRD + risk register reconciliation v1.1	Patched four assumptions (A6, A7, A9, A10) missing from the formal register; added the missing Owner column to the risk register.